Linear systems and elimination

# Sharp row-deletion singular-value limit for spherical random matrices

**Difficulty:** challenging **Importance:** interesting to the community

**Status: Open**

**Rating rationale:** Challenging reflects a sharp limit for the least singular value after adversarial row deletion; community impact includes random matrix theory and robust iterative solvers.

## Problem statement

Fix $0 < \theta < 1$. For $A \in \mathbb{R}^{m\times n}$ define

$$s_\theta(A) = \min_{S\subseteq\{1,\ldots,m\},\ |S|=\lfloor\theta m\rfloor} \min_{\|x\|_2=1} \|A_S x\|_2,$$

where $A_S$ contains the rows indexed by $S$. Let $a_\theta > 0$ satisfy $\mathbb{P}(|G| \le a_\theta) = \theta$ for $G \sim N(0,1)$, and set

$$h_\theta = \frac{1}{\sqrt{2\pi}} \int_{-a_\theta}^{a_\theta} t^2 e^{-t^2/2}\, dt.$$

For every integer sequence $(m_j, n_j)$ with $n_j \to \infty$ and $m_j/n_j \to \infty$, let the rows of $A_j \in \mathbb{R}^{m_j\times n_j}$ be independent and uniformly distributed on $\mathbb{S}^{n_j-1}$. Is

$$\frac{s_\theta(A_j)^2}{\|A_j\|_2^2} \xrightarrow{\mathbb{P}} h_\theta?$$

This states Steinerberger's proposed asymptotic with $\theta = q - \beta$. Convergence in probability and the floor convention make the source's limiting statement explicit. The source also asks for quantitative error bounds; these belong to this problem rather than separate entries.

## Connection to numerical linear algebra

The quantity measures the worst conditioning remaining after rows are discarded. It controls convergence guarantees for quantile Kaczmarz methods on corrupted linear systems.

### References

1. S. Steinerberger, *Quantile-based Random Kaczmarz for corrupted linear systems of equations*, Information and Inference **12**(1) (2023), 448–465, §2.3, equation (8) and the following paragraph. Published paper. Author preprint, v1 (2021), §2.3, equation (⋄).
2. E. Battaglia, J.-F. Cai, J. Chen, A. Ma, D. Needell, and T. Wu, *Quantile Randomized Kaczmarz for Streaming Linear Systems with Massart Noise*, arXiv:2608.27968v1 (2026), §1.1, discussion of Steinerberger's random-matrix heuristic and the distinction between static and streaming data; §5. Preprint.

### Earlier status check — 2026-09-08

Checked the published formulation and the arXiv version history (2107.05554 has only v1), then searched the problem's title, the author's name with "subsingular", and combinations of "trimmed/smallest singular value", "quantile", "Kaczmarz", "heuristic", "proof", "2025", and "2026". The August 2026 streaming paper still describes the original estimate as a heuristic; its fresh-sample convergence bounds do not establish this static-matrix limit. No later proof or counterexample was located in this bounded search. That is a literature-status assessment, not a proof of openness.

### Audit update — 2026-09-10

Rechecked Steinerberger's primary formulation, §2.3, and the 2026 streaming follow-up. Searches found no proof of the static proportional-deletion limit. Cai, Chen, Ma, and Wu's new subsample-size theorem, SIAM J. Matrix Anal. Appl. 47 (2026), 802–823, concerns quantile estimation during QRK, a different asymptotic quantity.